

WhiteboardOS — Master Technical Submission

Documentation

AI-Powered Multimodal Sketch-to-Code Living Prototype Engine

LEAD ARCHITECT: JAGADEESH CHINTA (90%) | TRACK: MULTIMODAL AI | PRODUCTION v1.0

Live Deployed App (Vercel)	https://sketch2-code-yd7c-git-master-jc-project.vercel.app
GitHub Repository	https://github.com/jagadeeshchinta/Sketch2Code
AI Evaluation Suite	https://sketch2-code-yd7c-git-master-jc-project.vercel.app/eval
Optical Camera Studio	https://sketch2-code-yd7c-git-master-jc-project.vercel.app/create/live
Core Technology Stack	Next.js 15 App Router, TypeScript 5, Gemini Vision API, TailwindCSS, SSE

1. What We Built and How It Works

A. The Core Problem

Translating rough whiteboard sketches, paper wireframes, and conversational voice ideas into clickable, production-ready frontend code takes 3 to 5 business days of manual Figma prototyping and frontend coding. This creates severe design-to-code friction, boilerplate waste, and context loss.

B. The Solution & End-to-End Pipeline

WhiteboardOS is an autonomous multimodal generative compiler that converts hand-drawn paper sketches, live optical camera scans, digital drawings, and spoken voice blueprints into functional, responsive, interactive web prototypes in under 10 seconds.

- 1. Multimodal Ingestion Layer:** Voice-to-UI dictation (Web Speech API), WebRTC Live Camera scanner, File dropzone, and HTML5 digital canvas board.
- 2. Client Image Pre-Processor:** Downscales smartphone photos to 1024px WebP (92% bandwidth reduction) and applies adaptive contrast stretching for faint pencil markings.
- 3. Spatial & Semantic Vision AI:** Extracts 30+ UI component types with 2D spatial bounding boxes, OCR text transcription, and confidence scores (averaging 96%).
- 4. Real-Time SSE Code Synthesis:** Streams production HTML5, CSS, and React code line-by-line via Server-Sent Events into an animated macOS developer terminal HUD.
- 5. Sandboxed Living Prototype:** Renders live clickable prototypes in an isolated iframe with natural language conversational iteration and 1-click ZIP export.

2. Why This Could NOT Have Been Built 2 Years Ago

1. Zero-Shot Multimodal Spatial Parsing (2022 vs 2026)

2 Years Ago: Visual LLMs did not exist or were restricted to basic image captioning ("A hand drawing on paper"). Extracting nested bounding boxes for 30+ distinct UI components required custom YOLO models.

Today: Gemini Multimodal Vision natively performs sub-pixel spatial coordinate bounding, layout hierarchy deduction, and handwritten OCR transcription in a single zero-shot forward pass.

2. Sub-Second Real-Time Token Streaming (SSE)

2 Years Ago: Legacy visual inference required 30–60 seconds per visual call, making interactive real-time prototyping unfeasible.

Today: Multi-tier model cascading and Server-Sent Events (SSE) stream code tokens with first-token latency under 800ms and total synthesis in ~8.4 seconds.

3. Strict Deterministic JSON Schema Enforcement

2 Years Ago: Legacy LLMs frequently broke output formats with markdown preambles and unescaped strings, causing JSON parse crashes.

Today: Native responseMimeTypes: "application/json" guarantees strictly typed, deterministic JSON schemas for reliable automated AST compilation.

3. My Engineering Innovations (What I Built vs. Thin Wrappers)

1. Multi-Model & Multi-Key Cascade Gateway

Built automated failover across gemini-3.5-flash, gemini-3.6-flash, and gemini-flash-latest with multi-key rotation to eliminate HTTP 429 quota exhaustion during live judging.

2. In-Browser Offline Heuristic CV Engine

Built an HTML5 Canvas spatial density analyzer and offline heuristic compiler that produces working prototypes in 42ms even if network connectivity is 100% severed.

3. AST DOM Self-Healing Guardrails (guardrails.ts)

Developed an AST validator that inspects LLM output, auto-repairs unclosed HTML tags, injects responsive viewport tags, and strips malicious scripts.

4. Client-Side Contrast & WebP Pre-Processor

Built an automated canvas filter that downscales 4K mobile photos to 1024px WebP and applies adaptive contrast stretching for faint pencil sketches.

5. Real-Time SSE Streaming Pipeline

Implemented bidirectional Server-Sent Events delivering token streams directly into a syntax-highlighted macOS terminal HUD.

6. Liquid Glass Design Token Architecture

Crafted custom CSS tokens, frosted glass backdrop filters, specular ridge lighting, and Three.js 3D WebGL backgrounds (Silk & MoltenMetal).

4. Team Roles & Contribution Breakdown (90% / 10%)

Jagadeesh Chinta (Lead Full-Stack AI Architect) [90% Ownership]	- Designed complete full-stack architecture, Next.js 15 App Router structure, and API routes. - Implemented Gemini Multimodal Vision integration, OCR extraction, and domain classification. - Built the multi-model cascade gateway, key rotation, and in-browser offline heuristic CV engine. - Engineered SSE streaming pipeline (/api/generate/stream), macOS terminal HUD, and AST guardrails. - Crafted Liquid Glass design token system, WebGL shaders, and Vercel serverless optimizations.
Teammate (Frontend UI & QA Assistant) [10% Ownership]	- Wireframe Asset Sourcing: Collected and cataloged sample wireframe sketches. - QA Benchmark Testing: Assisted in running test cases through the /eval suite. - Cross-Browser Verification: Tested UI across Chrome, Edge, and mobile Safari.

5. Key Technical Decisions & Challenges Solved

Challenge 1: Resolving LLM Domain Bias (Menus vs. Generic Dashboards) Problem: Generic vision prompts produced repetitive SaaS metrics cards even when given restaurant menus or e-commerce wireframes. Solution: Re-engineered ANALYSIS_SYSTEM_PROMPT with strict OCR extraction and domain routing rules, ensuring sketches with "MENU", "HOURS", "LOCATION", or "PRICE" synthesize authentic restaurant websites with real dish cards and reservation forms.
Challenge 2: Vercel Serverless Function 10s Timeouts Problem: Deep multimodal vision analysis and full-page code synthesis required 15–20s, exceeding Vercel's default 10s timeout. Solution: Configured export const maxDuration = 60 and export const dynamic = "force-dynamic" across all Next.js App Router API routes.

6. Evaluation Benchmarks & Telemetry (Live at /eval)

Evaluation Metric	Measured Score	Benchmark Target	Status
Component Detection Recall	96.2%	> 85% required	PASSED
End-to-End Latency	8.4 seconds	< 15 seconds target	PASSED
First-Token Streaming Latency	780 ms	< 2,000 ms target	PASSED
Average Token Cost	\$0.00018 / run	< \$0.01 / run budget	PASSED
Daily Cost (1,000 runs)	\$0.18 / day	< \$1.00 / day ceiling	PASSED
Offline Resilience Fallback	100% Functional	Fallback Required	PASSED (42ms)

